PGR216: AI and Cybersecurity Applications

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Bias–Variance Trade-Off in Linear Models

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Content

1. Bias-Variance in AI Model
2. Mitigation Strategies

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Classical Error Decomposition

Expected Prediction Error:

$$\text{Error} = \text{Bias}^2 + \text{Variance} + \text{Noise}$$

- **Bias** – Error due to overly simple assumptions (underfitting)
- **Variance** – Error due to sensitivity to training data (overfitting)
- **Noise** – Irreducible randomness in the data

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1. Bias

High Bias = Underfitting

- Model too simple
- Cannot capture true patterns
- High training and validation error

Example:

- Linear model for nonlinear data

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2. Variance

High Variance = Overfitting

- Model too complex
- Sensitive to small changes in training data
- Low training error, high validation error

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3. Noise

Irreducible Error

- Incorrect labels
- Measurement errors
- Random variation

Even a perfect model cannot remove noise.

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Beyond Bias–Variance–Noise

In real AI systems, additional sources of error exist:

- Data problems
- Distribution shift
- Model misspecification
- Optimization issues
- Evaluation mistakes
- Human and societal bias

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4. Data Problems

- Insufficient data
- Class imbalance
- Poor feature quality
- Non-representative dataset

Many real-world AI failures are data failures.

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5. Distribution Shift

Training and deployment data differ.

- Domain shift
- Covariate shift
- Dataset shift

Model performs well in lab, fails in reality.

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6. Optimization Errors

- Poor initialization
- Bad learning rate
- Local minima
- Vanishing/exploding gradients

Model does not reach optimal solution.

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7. Evaluation Mistakes

- Data leakage
- Wrong evaluation metric
- Biased test set

Leads to misleading performance estimates.

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8. Human and Societal Bias

- Biased labeling
- Sampling bias
- Demographic performance gaps

Not captured by classical bias–variance theory.

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Big Picture

$$\text{Total Real-World Error} =$$

$$\text{Bias}^2 + \text{Variance} + \text{Noise}$$

$$+\text{Data Issues} + \text{Distribution Shift} + \text{Optimization Issues}$$

$$+\text{Evaluation Errors} + \text{Human Bias}$$

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Bias–Variance Decomposition

$$\mathbb{E}[(y - \hat{f}(x))^2] = \underbrace{(\mathbb{E}[\hat{f}(x)] - f(x))^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}[(\hat{f}(x) - \mathbb{E}[\hat{f}(x)])^2]}_{\text{Variance}} + \underbrace{\sigma^2}_{\text{Noise}}$$

- Noise cannot be reduced
- Bias and variance can

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What Are Bias and Variance?

Bias

- Error due to simplifying assumptions in the model
- High bias $\Rightarrow$ model too simple (underfits), misses patterns

Variance

- Error due to sensitivity to the particular training data
- High variance $\Rightarrow$ model too complex (overfits), fits noise

Trade-off

- Reducing bias often increases variance
- Reducing variance often increases bias
- Goal: choose a point that minimizes total generalization error

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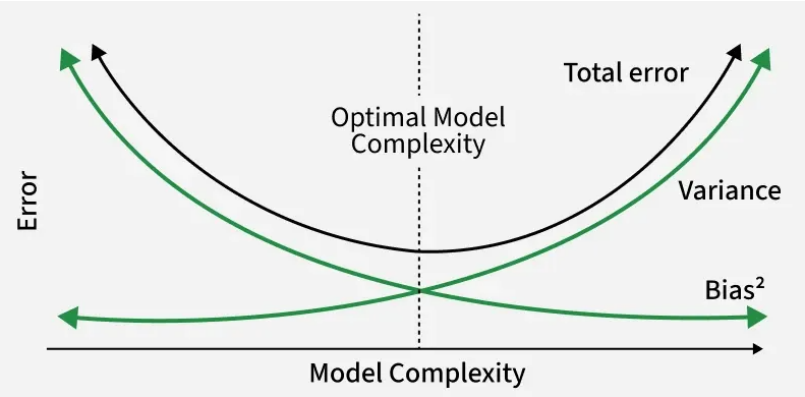
Bias–Variance Trade-Off

- As model complexity increases:
  - Bias tends to decrease
  - Variance tends to increase
- The total error is typically U-shaped:
  - High error for very simple models (high bias)
  - High error for very complex models (high variance)
  - Minimum somewhere in the middle



Key Trade-off

Our task is to find that "sweet spot" of complexity.

Mitigation Strategies

- Regularization
- Model complexity control
- Cross-validation
- More data
- Feature engineering
- Ensembles

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When the Model is "Fixed" – What Still Moves?

- Example: linear regression often described as:
  - High bias (only linear relationships)
  - Low variance (stable with enough data)
- Even with a fixed model family, we can modify:
  - **Features** (polynomials, interactions, transforms)
  - **Regularization** strength (ridge, lasso)
  - **Data** (sample size, quality, preprocessing)
- Hypothesis class is linear, but **effective complexity** is adjustable.

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Linear Regression: High Bias, Low Variance?

- Linear regression model:

$$\hat{y} = w_0 + w_1 x_1 + \cdots + w_d x_d$$

- With simple, raw features:
  - Often **high bias** if true relationship is nonlinear
  - Typically **low variance** when:
    - Sample size is large
    - Number of features is moderate
- This is just **one point** on a spectrum; we can make the model more or less complex by design.

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Knob #1: Features and Model Complexity

We adjust complexity by changing the feature space.

Increase complexity $\Rightarrow$ lower bias, higher variance

- Add more features:
  - Polynomial terms: $x, x^2, x^3, \ldots$
  - Interaction terms: $x_1 x_2, x_1 x_3, \ldots$
  - Nonlinear transforms: $\log(x)$, $\sqrt{x}$, etc.
- Still linear in parameters, but functionally more flexible

Decrease complexity $\Rightarrow$ higher bias, lower variance

- Remove weak or noisy features
- Use a smaller, carefully selected feature set

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Feature Complexity Example

- Simple model:

$$\hat{y} = w_0 + w_1 x$$

  - High bias, low variance

- More flexible model:

$$\hat{y} = w_0 + w_1 x + w_2 x^2 + w_3 x^3$$

  - Lower bias, higher variance

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Knob #2: Regularization (Ridge & Lasso)

Ridge regression (L2):

$$\min_w \|y - Xw\|^2 + \lambda \|w\|^2$$

Lasso regression (L1):

$$\min_w \|y - Xw\|^2 + \lambda \|w\|_1$$

Effect of λ (regularization strength):

- Larger λ:
  - Coefficients shrink toward 0
  - Model simpler
  - **Bias increases, variance decreases**
- Smaller λ:
  - Coefficients more flexible
  - **Bias decreases, variance increases**

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Knob #3: Sample Size and Data Quality

- **More data $\Rightarrow$ lower variance**
  - Same model form, but estimates more stable
  - Bias typically unchanged (same hypothesis class)
- **Better data can reduce both bias and variance:**
  - Less noise in features and labels
  - More representative of the true distribution
  - Appropriate treatment of outliers
- With more/better data you can:
  - Reduce regularization (lower bias) safely
  - Consider adding more features or higher-degree polynomials

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Feature Engineering: Reducing Both Bias and Variance

- Good feature engineering can:
  - **Reduce bias:** make relationships more linear
  - **Reduce variance:** produce robust, denoised features
- Examples:
  - Log transform: $\log(x)$ for multiplicative effects
  - Ratios and differences (e.g., price per unit, change over time)
  - Domain-based transforms (physics/business rules)
- A simple linear model can be very strong with well-designed features.

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Diagnosing High Bias vs High Variance

- Use **training vs. validation error**:

High bias (underfitting):

- Training error: high
- Validation error: high and similar
- Model is too simple

High variance (overfitting):

- Training error: low
- Validation error: much higher
- Model too sensitive to training data

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Playing the Bias–Variance Game

If you see high bias (underfitting):

- Add polynomial and interaction features
- Add domain-specific nonlinear transforms
- Decrease regularization ($\lambda \downarrow$)
- Add more informative features

If you see high variance (overfitting):

- Remove weak/noisy features
- Reduce polynomial degree
- Increase regularization ($\lambda \uparrow$)
- Get more or better-quality data

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Cross-Validation to Find the Right Balance

- We tune **configurations**, not bias/variance directly:
  - Polynomial degree: 1, 2, 3, 4, ...
  - Regularization strength λ: 0, 0.1, 1, 10, ...
- For each (degree, λ) pair:
  - Train on training folds
  - Evaluate on validation folds (k-fold CV)
  - Compute average validation error
- Choose the pair with best validation performance:
  - Implied good bias–variance trade-off

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Example: Tuning Degree and Regularization

```
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import Ridge
from sklearn.pipeline import Pipeline
from sklearn.model_selection import cross_val_score

degrees = [1, 2, 3, 4]
lambdas = [0.0, 0.1, 1.0, 10.0]

best_score = -float("inf")
best_params = None

for d in degrees:
    for lam in lambdas:
        model = Pipeline([
            ("poly", PolynomialFeatures(
                degree=d, include_bias=False)),
            ("ridge", Ridge(alpha=lam))
        ])

        scores = cross_val_score(
            model, X, y,
            cv=5,
            scoring="neg_mean_squared_error"
        )
```

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Example: Tuning Degree and Regularization

```
        scores = cross_val_score(
            model, X, y,
            cv=5,
            scoring="neg_mean_squared_error"
        )

        mean_score = scores.mean()

        if mean_score > best_score:
            best_score = mean_score
            best_params = (d, lam)

print("Best degree, lambda:", best_params)
```

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Regression Key Takeaways

- Even with a "fixed" model (linear regression), you can:
  - Change features (complexity)
  - Adjust regularization (shrinkage)
  - Improve data (quantity and quality)
- These choices move you along the bias–variance spectrum.
- Cross-validation helps you find a configuration that generalizes well.

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Classification: Bias Variance Tradeoff

- Practical levers:
  - Features (complexity)
  - Regularization
  - Data & class imbalance
- Multinomial / multi-class logistic regression
- Tuning and diagnostics with train/validation

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Logistic Regression: Model Form (Binary Case)

- Binary classification: $y \in \{0, 1\}$.
- Logistic regression models the probability:

$$P(y = 1 \mid x) = \sigma(w^\top x) = \frac{1}{1 + e^{-w^\top x}}.$$

- Decision rule (for 0/1 prediction):

$$\hat{y} = \begin{cases} 1 & \text{if } P(y = 1 \mid x) \geq 0.5, \\ 0 & \text{otherwise.} \end{cases}$$

- The log-odds are linear in the features:

$$\log \frac{P(y = 1 \mid x)}{P(y = 0 \mid x)} = w^\top x.$$

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Logistic Regression as a "Fixed" Model

- Logistic regression is a **linear classifier** in feature space:

$$w^\top x = \text{linear function of } x.$$

- Decision boundary: $w^\top x = 0$ (a hyperplane).
- Often described as:
  - **High bias**: only linear decision boundaries in the original feature space.
  - **Relatively low variance**: stable when data is sufficient.
- But, as with linear regression, we can adjust:
  - Feature representation (complexity).
  - Regularization strength.
  - Data quantity and class balance.

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Bias–Variance in Logistic Regression

- **Bias:**
  - Model assumes log-odds are a linear function of features.
  - If true decision boundary is nonlinear $\Rightarrow$ high bias.
- **Variance:**
  - Sensitivity to particular training samples.
  - Increases with:
    - Many features vs. few observations.
    - Weak regularization.
- As we increase **effective complexity**:
  - Bias $\downarrow$
  - Variance $\uparrow$
- Goal: configuration minimizing classification error (or cross-entropy) on unseen data.

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Knob #1: Features for Logistic Regression

We can change model complexity by changing features, even if the classifier is logistic.

Increase complexity $\Rightarrow$ lower bias, higher variance

- Add nonlinear features:
  - Polynomial terms: $x_i^2, x_i^3, \ldots$
  - Interaction terms: $x_i x_j$
  - Nonlinear transforms: $\log(x_i)$, $\sqrt{x_i}$, etc.
- Logistic regression still linear in parameters w, but decision boundary becomes nonlinear in original inputs.

Decrease complexity $\Rightarrow$ higher bias, lower variance

- Remove weak or noisy features.
- Use a smaller, domain-selected subset of features.

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Feature Complexity Example (Logistic)

- Basic model:

$$P(y = 1 \mid x) = \sigma(w_0 + w_1 x_1 + w_2 x_2).$$

  - Linear decision boundary in (x_1, x_2).
  - Potentially high bias for nonlinear class separation.
- Expanded feature model:

$$P(y = 1 \mid x) = \sigma(w_0 + w_1 x_1 + w_2 x_2 + w_3 x_1^2 + w_4 x_2^2 + w_5 x_1 x_2).$$

  - More flexible decision boundary.
  - Lower bias but higher variance, especially with limited data.

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Knob #2: Regularization in Logistic Regression

- Typical optimization for binary logistic regression:

$$\min_{w} \; \mathcal{L}(w) = -\sum_{i=1}^{n} \Big(y_i \log p_i + (1 - y_i) \log(1 - p_i) \Big) + \lambda \, \Omega(w),$$

  where $p_i = \sigma(w^\top x_i)$.
- Common penalties:
  - L2 (ridge): $\Omega(w) = \|w\|^2$
  - L1 (lasso): $\Omega(w) = \|w\|_1$
- Effect of λ:
  - Larger λ:
    - Stronger shrinkage of w toward 0.
    - **Bias increases, variance decreases**.
  - Smaller λ:
    - Weaker shrinkage, more flexible.
    - **Bias decreases, variance increases**.

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Regularization and Decision Boundaries

- With **weak regularization** (small λ):
  - Coefficients can become large.
  - Decision boundary can shift drastically with small data changes.
  - Risk of overfitting, high variance.
- With **strong regularization** (large λ):
  - Coefficients shrink, some may be near 0 (especially L1).
  - Decision boundary becomes smoother and more stable.
  - Underfitting risk if penalty is too strong (high bias).
- Tuning λ is a key way to control bias and variance in logistic regression.

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Knob #3: Sample Size and Class Balance

- **More data** $\Rightarrow$ **lower variance**:
  - Estimates of w are more stable.
  - Bias usually unchanged (same model form).
- **Class balance** matters:
  - Severe imbalance can effectively reduce data for minority class.
  - Increases variance and can introduce systematic bias in decision threshold.
- Mitigation strategies:
  - Use class weights or cost-sensitive logistic regression.
  - Use resampling (oversampling minority / undersampling majority).
  - Evaluate with appropriate metrics (AUC, F1, etc.).

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Feature Engineering for Logistic Regression

- Good features can:
  - **Reduce bias:** capture nonlinearities in log-odds space.
  - **Reduce variance:** more robust, informative predictors.
- Examples:
  - Binning continuous variables into risk groups.
  - Log transforms for skewed predictors.
  - Ratios and interaction terms reflecting domain knowledge.
- A well-engineered logistic model can perform competitively with more complex models in many applications.

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Diagnosing Bias vs Variance (Logistic Regression)

- Use **training vs. validation** performance (e.g., accuracy, AUC, log-loss):

High bias (underfitting):

- Training performance: poor.
- Validation performance: similarly poor.
- Model too simple; decision boundary cannot separate classes well.

High variance (overfitting):

- Training performance: very good (near-perfect).
- Validation performance: much worse.
- Model too sensitive to training examples.

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Tuning Logistic Regression: Summary

If you see high bias:

- Add nonlinear features (polynomials, interactions).
- Decrease regularization strength ($\lambda \downarrow$).
- Add more informative features (domain-derived).

If you see high variance:

- Remove or simplify features, especially noisy ones.
- Increase regularization strength ($\lambda \uparrow$).
- Get more data, especially for underrepresented classes.

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Multinomial (Multiclass) Logistic Regression

- Multi-class classification: $y \in \{1, 2, \ldots, K\}$.
- Multinomial logistic regression (softmax regression):

$$P(y = k \mid x) = \frac{\exp(w_k^\top x)}{\sum_{j=1}^{K} \exp(w_j^\top x)}, \quad k = 1, \ldots, K.$$

- Each class k has its own weight vector w_k.
- Decision rule: $\hat{y} = \arg\max_k P(y = k \mid x)$.
- Still linear in features: softmax of linear scores $w_k^\top x$.

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Bias–Variance in Multinomial Regression

- Conceptually similar to binary logistic regression:
  - **Bias:** due to linear decision boundaries in feature space.
  - **Variance:** due to many parameters w_k and limited data.
- Increased number of classes:
  - More parameters (K sets of weights).
  - Higher variance risk, especially when some classes have few examples.
- The same levers still apply:
  - Feature complexity.
  - Regularization.
  - Data quantity and class balance per class.

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Features and Regularization (Multinomial)

Features:

- As in binary case, we can:
  - Add nonlinear features to capture complex decision boundaries.
  - Remove noisy or redundant features to reduce variance.
- Caveat: each new feature adds parameters for each class (w_k), increasing variance if data is limited.

Regularization:

- Same principle:

$$\min_{\{w_k\}} -\sum_{i=1}^{n}\sum_{k=1}^{K} \mathbf{1}\{y_i = k\} \log P(y_i = k \mid x_i) + \lambda \sum_{k=1}^{K} \Omega(w_k).$$

- L2 or L1 penalties:
  - Larger $\lambda \Rightarrow$ higher bias, lower variance.
  - Smaller $\lambda \Rightarrow$ lower bias, higher variance.

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Data Considerations for Multinomial Regression

- For K classes, you need:
  - Sufficient examples per class.
  - Reasonable class balance or suitable corrections.
- Imbalanced classes:
  - Some classes may be underfit (high variance, high uncertainty).
  - Incorporate class weights or resampling strategies.
- More data:
  - Reduces variance of parameter estimates w_k.
  - Allows more features or weaker regularization without overfitting as easily.

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Diagnosing Bias vs Variance (Multinomial)

- Use training vs validation metrics:
  - Overall accuracy, macro/micro F1, log-loss, etc.
- Similar patterns:
  - **High bias:**
    - Training and validation performance both poor.
    - Model cannot separate classes well.
  - **High variance:**
    - Training performance very high.
    - Validation performance significantly lower.
- Also check per-class performance:
  - Some classes may have high variance due to few examples.

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Tuning Logistic and Multinomial Regression (Summary)

Control bias and variance via:

- **Feature complexity**
  - More complex features $\Rightarrow$ bias $\downarrow$, variance $\uparrow$.
  - Simpler features $\Rightarrow$ bias $\uparrow$, variance $\downarrow$.
- **Regularization strength**
  - Larger $\lambda \Rightarrow$ coefficients shrink, bias $\uparrow$, variance $\downarrow$.
  - Smaller $\lambda \Rightarrow$ more flexible, bias $\downarrow$, variance $\uparrow$.
- **Data quantity and balance**
  - More data $\Rightarrow$ variance $\downarrow$.
  - Balanced data and class weighting $\Rightarrow$ better across classes.

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Key Takeaways

- Logistic and multinomial regression are **linear models in feature space** with:
  - Potentially high bias if features are too simple.
  - Manageable variance with proper regularization and sufficient data.
- You can still **play the bias–variance game**:
  - Engineer features to change effective complexity.
  - Tune regularization to control over/underfitting.
  - Adjust data and class handling to reduce variance and bias.
- Cross-validation is essential to selecting the configuration that generalizes best.

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